

When Does Quantum Coherence Help Sequence Modeling? A Controlled Study

A task–resource boundary, characterized at three levels: task performance, generation quality, and the quantum-information content of the model’s state.

Abstract

We ask whether the resources of quantum mechanics — complex probability amplitudes, superposition, the Born rule, entanglement, and lossless unitary evolution — provide a *scaling* advantage when used as the computational substrate of an autoregressive sequence model, and if so, *when*. Building on a quantum-channel language model (QCLM) whose sequential engine is a quantum channel rather than attention or gated recurrence, we run a controlled experimental ladder in which each rung isolates a single quantum resource and each rung carries a *decoherence ablation* that collapses the quantum mechanism to a dimension-matched classical control. We find a sharp, reproducible boundary. On a synthetic long-range recall task, lossless unitary phase memory yields an advantage that **grows with both recall distance and state size**, and is destroyed by any mid-sequence measurement — the advantage is precisely the coherent phase. On natural language, the *same* mechanisms provide advantages that are real but **bounded (~0.1–0.16 bits) and, for lossless phase memory, net-negative** — confirmed across two token-embedding schemes, two metric families (perplexity and long-range generation coherence), and a quantum- information diagnostic. That diagnostic, adapted from the out-of-time-order- correlator / mutual-information framework of Singh et al. (2026), shows the model develops **~2.9 nats of cut-invariant mutual information** in its memory state while reading language, **~1 nat of it phase-attributable** — substantial, genuinely quantum correlation that the language task nonetheless does not convert into predictive advantage. The boundary is the result: quantum correlation helps when the task structurally requires lossless long-range structured memory, and is *built but unrewarded* when the task rewards real-valued statistical association, which attention already captures.

1. Introduction

Modern language models are built from self-attention (transformers) or gated / linear recurrence (RNNs, S4/Mamba). A different question motivates this work: can quantum mathematics — Hilbert spaces, complex amplitudes, the Born rule, quantum channels, and unitary evolution — serve *directly* as the engine of an autoregressive model, and does the “quantumness” pay for itself as the model scales?

The honest answer we reach is conditional, and the conditionality is the contribution. Prior “quantum machine learning” results frequently report a quantum model matching or beating a classical baseline on some task, leaving open *why*, and whether the advantage would survive scaling or a fair ablation. We instead fix a single architecture family, vary one quantum resource at a time, and attach to every comparison a decoherence ablation that yields an exactly dimension- and parameter-matched classical control. This lets us ask not “does it work?” but “**does the quantum resource scale, and on which tasks?**” — and to answer at three levels of analysis.

Contributions

1. **A controlled experimental ladder** isolating coherence, quantum routing, entanglement, and lossless unitary memory, each with a decoherence ablation, tested on both synthetic recall and natural language.
2. **A positive scaling result:** lossless unitary phase memory yields a recall advantage that grows with recall distance and with state size, and is destroyed by mid-sequence measurement — isolating coherent phase as the mechanism.
3. **A robust negative result for language:** the same resources are bounded (~ 0.1 – 0.16 bits) or net-negative, confirmed across embeddings, perplexity, and long-range generation-coherence metrics.
4. **A quantum-information diagnostic** (adapting the OTOC / mutual-information framework of Singh et al., 2026, to a non-RBM architecture) showing the model develops substantial, cut-invariant quantum correlation while modeling language — establishing that the bounded advantage is *not* a failure to build quantum structure but a property of the task–resource match.

2. The architecture family

The base object is a Quantum Channel Language Model: the context is carried as a quantum state, each token defines a quantum operation, and the next-token distribution is read out by the Born rule. We summarize the variants used as rungs; full definitions and the global-POVM constraint are in the appendix.

- **Pure QCLM.** State $\rho \in \mathbb{C}^{n \times n}$ (density matrix); each token owns Kraus operators defining a channel $E_x(\rho) = \sum_w K \rho K^\dagger$ and a POVM element $M_x = \sum_w K^\dagger K$, with a single global completeness constraint $\sum_x M_x = I$. The autoregressive law is

$$p(x_t \mid x_{<t}) = \text{Tr}(\rho M_{x_t}), \quad \rho_t = E_{x_t}(\rho) / p(x_t).$$

A complex-valued, trace-preserving *linear* state-space model whose only nonlinearity is the Born readout. Decoherence ablation: zero the off-diagonals of ρ

each step $\rightarrow$ exactly a classical Hidden Markov Model.

- **Hybrid (quantum features / routing + attention).** The pure QCLM is a linear state-space model; its correlations decay and it cannot do long-range or conversational structure. The hybrid uses a quantum sub-layer as a local mixer alongside causal self-attention for nonlinear all-to-all routing.
- **Lossless unitary memory.** A token-conditioned unitary recurrence $\psi_t = U(x_t) \psi_{t-1}$, with $U(x)$ formed by the **Cayley transform** of a learned Hermitian generator (exactly unitary, differentiable, norm-preserving over hundreds of steps; chosen over `matrix_exp`, which was non-unitary on complex input, and over `eigh`, which was unstable). The state is **never measured mid-sequence**; nonlinearity comes from the readout and the attention layers. Decoherence ablation: zero the phase in the readout (and, for the diagnostic, in the state), collapsing interference.

All quantum sub-layers reuse a validated isometry/unitary parametrization. The decoherence ablation is the spine of every comparison: it removes exactly the quantum resource while holding dimension and parameter count fixed.

3. The positive result: lossless unitary memory scales on recall

We built a synthetic copy task: the model sees a symbol, then must reproduce it after a controlled *delay* of intervening symbols. This task structurally requires lossless storage of information across distance — the property unitary evolution provides natively. We compare, at matched parameter count ($\sim 20\text{K}$ – 330K): the lossless unitary model, its decohered (phase-zeroed) twin, and a GRU of equal state size.

3.1 The advantage grows with recall distance

Final recall accuracy (chance = 0.125), state size $n = 32$:

delay	unitary (lossless)	unitary decohered	GRU
10	0.999	0.740	0.968
20	0.880	0.558	0.730
40	0.659	0.391	0.452
80	0.562	0.294	0.323

The unitary advantage over the GRU widens monotonically with delay (0.03 $\rightarrow$ 0.15 $\rightarrow$ 0.21 $\rightarrow$ 0.24). At short range everyone succeeds; as the recall distance grows, the GRU decays toward chance while the lossless unitary model

degrades gracefully. The decohered twin is *worst* at every delay — removing phase drops the unitary model below even the classical GRU, so the advantage is unambiguously the coherent phase.

3.2 Mid-sequence measurement destroys the advantage

At fixed delay 40, sweeping how often the state is measured mid-sequence (`measure_every`):

measure_every	unitary accuracy
0 (lossless)	0.659
2	0.421
4	0.396
8	0.514

Any mid-sequence collapse degrades performance toward the classical floor (~ 0.52). There is no coherence-vs-nonlinearity “sweet spot” for recall: the task wants zero mid-sequence measurement. The nonlinearity must come from elsewhere (the readout, or — in a language model — the attention layers).

3.3 The advantage grows with state size

At fixed delay 40, sweeping state size n :

n	unitary (lossless)	decohered	GRU
32	0.659	0.391	0.558
64	0.894	0.622	0.480
128	0.869	0.778	0.509

The unitary model converts additional state capacity into recall — leaping 0.66 $\rightarrow$ 0.89 from $n=32$ to $n=64$ — while the GRU stays flat (~ 0.48 – 0.51) at any size. A unitary model at $n=64$ (83K params) beats a GRU at $n=128$ (300K params). The advantage grows on a second axis (state size), confirming §3.1.

Conclusion (positive): lossless unitary phase memory is a quantum resource whose advantage *grows* with the relevant axes (recall range and state size) and is destroyed by measurement — categorically different from the bounded resources below. On a task structurally built to need lossless structured memory, the quantum mechanism is the right tool and it scales.

4. The negative result: language does not reward the same resources

We then tested whether this transfers to natural language (character-level tiny-Shakespeare; the hybrid integrates the lossless memory track with attention + MLP blocks, reading the complex memory state non-destructively into the residual stream). All comparisons use best-validation checkpoints with dropout and weight decay (the lossless memory is a high-capacity memorizer and overfits a small corpus without regularization — itself indirect evidence the mechanism is powerful).

4.1 Perplexity: the decohered (classical) regime wins

Best validation bits/char (lower is better), 13.5M params, 3000 steps:

	real embedding	complex (amplitude+phase) embedding
intact (phase on)	2.250	2.232
decohered (phase off)	2.136	2.156

The decohered model wins under both embeddings. A learned complex amplitude+phase embedding — making phase a first-class property of the *data*, testing whether a classical lookup interface was starving the mechanism — improved the intact model only marginally ($2.250 \rightarrow 2.232$) and did not flip the sign. The interface was not the bottleneck. Across the earlier rungs the same pattern held: quantum coherence as features or as Born-rule routing contributed a *bounded* ~ 0.16 bits; entanglement on a tensor-product Hilbert space contributed a bounded ~ 0.10 bits that did not grow with the number of entangled subsystems.

4.2 Generation coherence: the decohered regime also wins

Perplexity scores local prediction and could miss long-range coherence, so we measured generation-level coherence over 40 samples, calibrated to the real corpus (speaker recurrence 0.972, real cast 172 characters):

config	val bits/char	speaker recurrence	real-cast fraction
complex intact	2.232	0.066	0.860
complex decohere	2.156	0.352	0.951
real intact	2.250	0.079	0.740
real decohere	2.136	0.250	0.915

The decohered model is more coherent on every axis — 4–5 $\times$ higher speaker recurrence (returning to an established cast rather than inventing new names) and

higher real-cast fraction. An initial qualitative impression that intact samples looked more coherent did not survive quantitative measurement; the coherence metrics point the *same* way as perplexity. Phase is neutral-to-harmful for language on both metric families.

5. The quantum-information level: the resource is built but unrewarded

To establish *why* the advantage is bounded, we adapt the OTOC / mutual-information diagnostic of Singh et al. (2026, §3.2) — which for RBMs relates the imaginary part of an out-of-time-order correlator to a subsystem covariance and bounds it against quantum mutual information — to our non-RBM architecture, estimating the quantities numerically from the model’s memory state. We treat the unitary memory state $\psi \in \mathbb{C}^{128}$ as a pure state on a bipartite Hilbert space and, as the model reads real text, compute the mutual information $I(A : B)$ and entanglement across the cut.

Mutual information (nats) across two independent bipartitions:

	16×8 cut	8×16 cut
complex intact	2.94	2.94
complex decohered	1.93	1.93
real intact	2.88	2.88
real decohered	2.04	2.02

Two facts matter. First, **MI is invariant to the cut** (stable to two decimals), so it is a genuine property of the state, not a bipartition artifact — the quantity is publishable. Second, the intact state develops **~2.9 nats of mutual information at ~0.5–0.7 entanglement saturation** — substantial, far from product-like — of which **~1 nat is phase-attributable** (intact minus decohered, 1.0 and 0.86 for the two models).

The interpretation is the crux. The model is *not* correlation-limited: it builds about a nat of phase-driven quantum correlation while reading language. Yet that correlation yields at most ~0.1 bit of predictive advantage, net-negative after generalization. The resource is present (measured, ~1 nat), it is used (half-saturation entanglement), and the language task does not convert it into predictive value — in direct contrast to the recall task, where the same kind of correlation scales with and drives performance.

(Methodological note: we report MI and entanglement, which are basis-invariant for a pure-state bipartition. An ℓ_1 -coherence-of- ρ_A measure and a computational-basis covariance η were also computed but did not cleanly isolate phase — a real ρ_A can carry large off-diagonals — and are not used in the claims. That the data exposed this is the diagnostic working as intended.)

6. The boundary, and what it means

Across three levels of analysis the same boundary appears:

- **Task performance:** quantum advantage *scales* on synthetic long-range recall (grows with delay and state size); is *bounded-to-negative* on language.
- **Generation quality:** the decohered regime is more coherent on language, not less — confirming the perplexity result on an independent metric family.
- **Quantum information:** the model develops substantial, cut-invariant, phase-attributable quantum correlation while modeling language — so the bounded advantage is a property of the *task-resource match*, not a failure to build quantum structure.

The boundary: quantum correlation helps when the task structurally requires *lossless long-range structured memory* (exact recall across distance), and is *built but unrewarded* when the task rewards *real-valued statistical association*, which attention already captures and which carries its useful information in similarity rather than phase.

This directly addresses the open problem stated in the QML-for-complex-systems literature — “a deeper theoretical understanding of when and why quantum resources improve learning performance.” The successful applications surveyed there are uniformly on quantum-structured data (many-body wavefunctions, molecular Hamiltonians, strongly-correlated sampling); natural language is not such a distribution, and our three-level characterization shows precisely how the mismatch manifests: the mechanism develops quantum correlation, the task does not pay for it.

Implications

For a **quantum-core language model**, the data are unambiguous: scale the decohered/classical regime (attention and depth do the conversational work; the quantum core is not the engine for language). For the **quantum machinery built here** — a differentiable, hardware-validated, token-conditioned unitary evolution engine with a clean decoherence ablation — the natural home is the domain where the surveyed literature finds genuine advantage: quantum-structured data (neural-network quantum states, molecular ground states, strongly-correlated sampling), where the recall-style scaling we demonstrated is expected to apply to the task it was built for.

7. Limitations and honest scope

- Language experiments are at character level on a ~1M-character corpus (tiny-Shakespeare). A large web corpus (FineWeb) with genuine long-range structure and unmemorizable scale *could* alter the language verdict; we judge this unlikely given the consistency across embeddings, metrics, and the MI diagnostic, but we did not run it, and a falsification test with the decohere twin as control remains the honest way to close the question definitively.
- The recall task is synthetic; it demonstrates the scaling mechanism cleanly but is not itself a language capability.
- The MI diagnostic is estimated numerically on a pure-state bipartition of the memory; it does not use the closed-form OTOC available only for RBMs, and the basis-dependent / coherence measures are reported but not relied upon.
- Throughput: the per-token unitary scan is sequential (~1600 tok/s eager at block 256; `torch.compile` was unstable on the target GPU). A parallel associative prefix-scan (unitary composition is associative) is the known fix required before any scale-up.

8. Reproducibility

Code: `qlm/unitary_hybrid.py` (lossless-memory + attention hybrid, complex-embedding option), `qlm/train_unitary_hybrid.py` (training with dropout / weight decay / best-VAL checkpointing), `qlm/recall_probe.py` (synthetic recall sweeps), `qlm/coherence_metric.py` (generation-coherence metrics), `qlm/otoc_diagnostic.py` (mutual-information diagnostic), and the base QCLM in `qlm/model.py`. Every quantum comparison has a `--decohere` control.

Appendix: the decoherence ablation as a controlled instrument

The single methodological commitment that makes this study interpretable is that each quantum resource is tested against a control that removes *only* that resource. For the density-matrix QCLM, zeroing off-diagonals collapses ρ to a classical HMM of identical dimension and parameter count. For the unitary memory, zeroing the readout/state phase removes interference while preserving amplitudes and parameters. Because the control is dimension- and parameter-matched, any performance difference is attributable to the quantum resource itself, not to capacity — and any *absence* of difference (as on language) is therefore meaningful rather than confounded.